

PROJECT PROPOSAL

1. Project Title

- SmartClassroom

2. Project Description

- **What the system does:** A web-based classroom application that handles role-based logins (Students, Teachers, and Admin), allows students to upload assignments, enables teachers to track submissions and grade them, and automates email notifications for assignment deadlines and results.
- **The purpose of the system:** To digitize and streamline the assignment submission, tracking, and grading process between students, teachers, and administrators in an educational institution.

3. Problem Statement

- Traditional methods of managing assignments via emails or physical submissions make it difficult to track who has completed or missed their work. Teachers face challenges in managing submissions and grading manually, while students often miss deadlines due to a lack of automated reminders. This system solves these issues through a centralized dashboard and automated tracking.

4. Project Objectives

- To develop a secure multi-role authentication system for Students, Teachers, and Admins.
- To provide a file upload and tracking mechanism for student assignments.
- To enable teachers to review submissions, track completed/pending students, and assign marks.
- To automate email notifications for assignment openings, deadlines, and results.
- To give administrators full control over user management (adding teachers and students).

5. Target Users

- **Students:** Login using Index Number and Password to upload assignments and receive email updates.
- **Teachers (Staff):** Login using Staff ID and Password to manage assignments, view submission statistics (who has done or not done), and grade work.
- **Administrators:** Login using Email and Password to manage the system, add teachers/students, and oversee overall progress.

6. Main System Features

- **Multi-Role Authentication:** Separate login systems for Admin, Teachers, and Students.
- **User Management (Admin):** Add and manage teachers and students.
- **Assignment Management:** Create, upload, and track assignments.
- **Submission Tracking:** Real-time visibility of who has uploaded or not uploaded assignments.
- **Grading & Reporting:** Teachers can check and record assignment marks.
- **Email Notification System:** Automated emails sent to students regarding assignment open times, finish times/deadlines, and results.

7. Proposed Technology Stack

- **Frontend:** React.js
- **Backend:** Fast API
- **Database:** MySQL

8. Expected Outcome

- A fully functional, responsive classroom web application that successfully automates assignment workflows, improves communication through automated emails, and ensures smooth administrative control over users and academic records.